

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Industry Problem Solving Task

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO4: Articulation (BL4, BL5)	Assessment:	Assessment Tool 2 – Industry Problem Solving Task
Weightage:	20% of CIA	Total Marks:	100
CO4 Statement:	Solve optimization problems using dynamic programming and greedy strategies.		
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Section / Batch:	B.Tech IT	Date:	22.08.2026

Instructions to Students

- Answer the following question. Total Marks: 100.
- Carefully analyze the given questions.
- Explain in detail with sample code if needed.
- Clearly identify all key issues.
- Provide a thorough analysis with validated results and performance evaluation.

Question Type	Focus Area	Questions
Industry Problem Solving Task	Focus on Solving optimization problems using dynamic programming and greedy strategies	Q1

SECTION A – Industry Problem Solving Task

Code of Conduct

I Ramya S S / 192421226 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Ramya S S

Q30. Video Streaming Server Allocation Optimization – Load Balancing

1. Problem Statement

A video streaming platform such as Netflix, YouTube or Amazon Prime Video must handle thousands or millions of users at the same time. User demand changes continuously, especially during live sports, movie releases or popular events. If too many users are assigned to one server, it may become overloaded, causing buffering and high response time. The objective is to distribute user requests among servers efficiently while maintaining low latency, sufficient bandwidth and high throughput.

2. DAA-Based Solution

This can be treated as an optimization and load balancing problem. For every incoming request, the system checks the available servers and selects the most suitable one based on:

- Current server load
- Available bandwidth
- Network latency
- Server capacity

A simple approach is:

1. Receive a video request from a user.
2. Check the available servers.
3. Remove servers that are overloaded or have insufficient bandwidth.
4. Calculate a cost for the remaining servers.
5. Select the server with the minimum cost.
6. Assign the user to that server.
7. Update the server's load.
8. If demand increases significantly, add more servers.

3. Comparison of Load Balancing Algorithms

Algorithm	Working	Advantages	Limitations
Round Robin	Sends requests sequentially to servers	Simple and fast	Does not consider actual server load
Least Connections	Selects server with minimum active connections	Reduces server overload	Does not consider bandwidth or latency
Least Load	Selects server with lowest current workload	Better resource utilization	Requires continuous monitoring
Dynamic Load Balancing	Considers load, bandwidth and latency	Suitable for streaming systems	More computation is required

For a large video streaming platform, Dynamic Load Balancing is more effective because server conditions change continuously.

4. Real-World Comparison

Consider a supermarket with multiple billing counters. If every customer is sent to Counter 1, a long queue will form while other counters remain free.

A smart manager sends new customers to a counter with a shorter queue.

Similarly:

Customer → Billing Counter

is similar to:

User → Streaming Server

The load balancer acts like the manager, selecting the best available server for every request.

5. Important Constraints

Bandwidth: High-quality 4K videos require more bandwidth than 480p videos. A server with insufficient bandwidth should not receive another high-quality stream.

Latency: The server with lower network delay should generally be preferred because it provides faster video loading.

Server Capacity: Each server can handle only a limited number of users. Overloading it can increase buffering and response time.

Traffic Spikes: During a cricket match or popular movie release, thousands of additional users may arrive within seconds. The system must scale servers dynamically.

6. Complexity Analysis

If every incoming request checks all n servers, the basic approach has:

Time Complexity = $O(m \times n)$

where:

- m = number of user requests
- n = number of servers

For example, with 10,000 requests and 100 servers, many server comparisons are required.

A Priority Queue or Min-Heap can be used to efficiently identify less-loaded servers and reduce the allocation overhead.

7. Performance Metrics

The algorithm can be evaluated using:

Metric	Expected Result
Response Time	Low
Throughput	High
Server Utilization	Balanced
Network Latency	Low
Buffering	Minimal
Scalability	High

For example, if Round Robin sends requests equally but one server has higher latency, a dynamic algorithm can choose another server and provide a better streaming experience.

8. Real-World Applications

This approach is useful in:

- Netflix and YouTube for video delivery
- Amazon Prime Video for streaming content
- Live sports platforms for sudden traffic increases
- CDNs for delivering videos from locations closer to users
- Cloud computing for distributing workloads
- Video conferencing where low latency is important

Conclusion

Dynamic load balancing is an effective DAA-based solution for video streaming server allocation. Unlike simple Round Robin, it considers server load, bandwidth, latency and capacity before allocating a request. Efficient algorithms and data structures such as greedy selection and priority queues can improve performance. The result is lower response time, higher throughput, balanced server utilization and reduced buffering, making the system suitable for real-world content delivery platforms.

RUBRICS FOR CALCULATION

Industry Problem Solving Task

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%				
Application of Engineering Concepts	25%				
Solution Design	25%				
Use of Tools	15%				
Analysis	10%				
Professionalism	5%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____